Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-5 — Completing the Square

Choose Your Route

Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.

Student name :	Date : .....

PRACTICE — Secure the method

Complete it, solve it, find the minimum

Items 1–3: complete the square. Items 4–6 are below.

1.	2.	3.
$x^2 + 16x + \square$	$x^2 - 5x + \square$	$x^2 + x + \square$

4. Solve $x^2 + 12x + 20 = 0$ by completing the square.

5. Solve $x^2 - 10x + 24 = 0$ by completing the square.

6. Find the minimum value of $f(x) = x^2 - 4x + 9$ and where it occurs.

Done when: the adjustment term is added to both sides and every solution is checked by substitution.

APPLY — Use it in context

A bigger coefficient, a jet and a rectangle

1. Solve $2x^2 + 8x - 5 = 0$. Give the exact solutions.

2. A model water jet has height $h(t) = -5t^2 + 20t + 1$ metres after t seconds. Find the greatest height and when it happens (model values).

3. A rectangular garden has perimeter 40 m. Write its area in terms of the width x, complete the square, and find the greatest area.

Done when: the extreme value and where it occurs are both stated with units.

INVESTIGATE — Find out why

Arguments, not answers

1. Why must the adjustment term be added to BOTH sides of the equation? Show with $x^2 + 6x = 7$.

2. Complete the square on $ax^2 + bx + c = 0$ and show that it gives the quadratic formula.

3. Complete the square on $y = ax^2 + bx + c$ to show that the vertex has $x = -\frac{b}{2a}$.

Done when: each argument keeps the equation balanced at every step.

Remember

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.